

Two Scales in a Convention-Locked Preferential-Attachment Tree: Fixed-Vertex Martingales and the Global Degree Profile

HCS-C321 theorem-and-evidence package

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Abstract

For the linear preferential-attachment tree started from one edge, with no self-loops, we join two scales in one boundary-complete theorem. Every fixed vertex has exact rising-factorial moments of all orders and a moment-determinate square-root limit, whereas every fixed degree population converges in L^2 to $4/[k(k+1)(k+2)]$. The proof keeps fixed-vertex degree and degree counts separate. An exact finite computation independently audits the recurrences but is not used to infer either asymptotic theorem.

1 Frozen process and exact local law

Let T_2 consist of the single edge $\{1, 2\}$. Given T_n , add vertex $n+1$ and connect it to exactly one old vertex v , chosen with probability

$$\Pr(v \mid T_n) = \frac{d_v(n)}{2(n-1)}.$$

Thus there are no loops or multiple edges, n is the vertex-count clock, and $\sum_{v \leq n} d_v(n) = 2(n-1)$. Write $D_i(n) = d_i(n)$ for one fixed vertex, $N_k(n) = \#\{v \leq n : d_v(n) = k\}$ for a population, and

$$s_i = \begin{cases} 2, & i = 1, 2, \\ i, & i \geq 3. \end{cases}$$

This precise convention matters: several standard formulations use a self-loop or a linearized-chord-diagram seed [1, 2]. Fixed-vertex limit lineage is discussed by Backhausz [3]; the constants below are nevertheless derived for the declared one-edge process.

For $x^{\bar{r}} = x(x+1) \cdots (x+r-1)$, the identity $(d+1)^{\bar{r}} - d^{\bar{r}} = rd^{\bar{r}}/d$ gives

$$\mathbb{E}[D_i(n+1)^{\bar{r}} \mid T_n] = \left(1 + \frac{r}{2(n-1)}\right) D_i(n)^{\bar{r}}. \quad (1)$$

Theorem 1 (Exact all-order degree moments). *For every integer $r \geq 1$ and $n \geq s_i$,*

$$\mathbb{E}D_i(n)^{\bar{r}} = r! \frac{\Gamma(n-1+r/2)}{\Gamma(n-1)} \frac{\Gamma(s_i-1)}{\Gamma(s_i-1+r/2)}. \quad (2)$$

Proof. Every vertex has degree one at its birth time, so its initial rising factorial is $r!$. Iterating (1) and converting the finite product to a gamma quotient proves (2). $\square$

Revision certificate: convention and exact factorial law

This original round freezes the one-edge model and proves every finite-time rising-factorial identity.

References

- [1] A.-L. Barabási and R. Albert, “Emergence of scaling in random networks,” *Science* 286 (1999), DOI: 10.1126/science.286.5439.509.
- [2] B. Bollobás, O. Riordan, J. Spencer, and G. Tusnády, “The degree sequence of a scale-free random graph process,” *Random Structures Algorithms* 18 (2001), DOI: 10.1002/rsa.1009.
- [3] Á. Backhausz, “Limit distribution of degrees in random family trees,” *Electron. Commun. Probab.* 16 (2011), DOI: 10.1214/ECP.v16-1598.